

DS 3500: Advanced Programming with Data

Fall 2026 Boston Campus

Course Information

Dates: Sep 9 – Dec 20, 2026

Lecture Schedule:

- [Sec 1] TF 1:35 PM - 3:15 PM (Mugar 201)
- [Sec 2] TF 3:25 PM - 5:05 PM (Churchill Hall 103)

Teaching Team:

Instructor

Deahan Yu (he/him)

Email

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Office hours / Location

Wed 10-11:30am / ME 316B
ME: Meserve Hall

Teaching Assistants

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Course Description

Offers a deep dive into the design and implementation of enterprise-grade software systems with an emphasis on software architectures for more complex data-driven applications. Covers extensible architectures that support testing, data provenance, reuse, maintainability, scalability, and robustness and building software APIs and libraries for wide-scale adoption and ease of use. Students design, implement, and test complex loosely coupled service-oriented architectures using distributed processing, stream-based data processing, and interprocess communication via message passing. Explores the features, capabilities, and underlying design of popular data analysis and visualization frameworks.

Course Prerequisites: DS 2500 or CS 2100

Course Learning Goals

When you complete this course, you should be well-positioned to contribute to large-scale, industry-level data engineering projects. Through active engagement and completion of course activities, you will be able to:

1. Use Git feature branches, meaningful commits, and GitHub to manage code the way industry teams do
2. Build config-driven cleaning pipelines that log every step so you can audit what happened to your data
3. Build production-ready FastAPI services with Pydantic validation, async endpoints, and proper HTTP status codes
4. Model and query relational data using SQLAlchemy, and swap between SQLite and PostgreSQL with a single config change
5. Develop an industry-level specialized system – an end-to-end ML platform or a scalable visualization platform.

Course Materials & Tools

You will need to have access to a device that connects to the internet so that you can access email, Canvas, Github. Students can borrow equipment and access other learning technology from [Information Technology Services](#). For tech support, see [ITS Northeastern Tech Resources](#).

Canvas: All course details and materials will be posted on our Canvas course site. You may also ask questions that may be helpful for your fellow classmates in Discussions.

Github: You will use your own Github account to work on all assignments (homeworks and project assignments). You are required to share your GitHub username with the teaching team. Once shared, your username should not be changed for the remainder of the semester. If you do change it for any reason, you must notify the teaching team immediately.

All of your course-related repositories must be set to *public* so that the teaching team can access and review them at any time until the last day of semester.

If the teaching team follows the link you provided and cannot locate your repository or work – for example, because your username changed without notice – it may affect your grade.

Share Your Github Username: <https://forms.gle/w3QiY2YsiMz1xYhA8>

Office hours

I will be holding regular office hours every Wednesday from 10:00–11:30 a.m. in Meserve Hall 316B. The room is on the 3rd floor—turn right when you reach the 3rd floor after taking the stairs or exiting the elevator.

If this time doesn't work for you and you'd like to schedule an appointment, you can fill out [the office-hour meeting request form](#) at any time. One of our TAs will reach out to you via email to arrange a meeting. Please note that if your preferred times are last-minute (e.g., within 48 hours of your request), we will do our best to accommodate your request, but we may not be able to do so.

If you'd like to talk to me outside of my office hours, feel free to reach out to me via email.

Course Expectations

Expectations for the course: Transparency & Progress over Perfection

We are here to guide your learning and will challenge you to actively engage in the learning process through class activities, assignments, and more. This course will provide clear and open communication regarding grading criteria, instructions, and policies. We will try our best to create a fair and trusting learning space so that you can fully understand what is expected and how to achieve your goals.

We believe in progress over perfection. This course is designed to encourage continuous effort for success and learning. We will do our best to give you the tools, feedback, and support to gradually develop your skills and knowledge, and will always welcome suggestions for improvement.

Expectation for all of us: Good Citizen

We are all citizens in this learning community, and it is important for us to show respect and courtesy towards one another. Let's be good citizens. As good citizens, we respect the perspective of people of all races, ethnicities, gender expressions and gender identities, religions, sexual orientations, disabilities, socioeconomic backgrounds, and nationalities.

Attendance

As an on-ground class, you are expected to attend in person, and attending lectures is highly encouraged since we do many hands-on programming exercises together. That said, attendance is not mandatory for lecture sessions, so there is no need to inform the teaching team about absences.

Quizzes and presentations will take place in class. These are assessments, not lectures, so you are expected to attend and complete them in class in order to receive grades. Make sure to mark your

calendars – missing an assessment will significantly affect your grade. See the Course Schedule for these dates.

Evaluation

Your grade will be evaluated based on factors below.

Factor	Points	Individual Assignment	Individual Assignment Points	Note
MP 1	1000 pts (10%)	-	-	
MP 2	1000 pts (10%)	-	-	
MP 3	1000 pts (10%)	-	-	
Final Project	1600 pts (16%)	Team Report	50 pts	
		Progress Report	100 pts	
		Demo Testing	100 pts	
		Demo Presentation	250 pts	
		Final Deliverable	300 pts	
		Peer Evaluation	800 pts	
Quiz	5400 pts (54%)	10 Quizzes	1450 pts	See Quiz section for more details
Total	10000 pts (100%)	-	-	

Letter Grades

To pass the course, you must attain a final grade of 60% (D-) or higher. Please note that we drop the decimal portion and keep only the whole number. So 94.01% becomes 94% but 93.99% becomes 93%. Your final grade for the course will use the following breakpoints to convert from letter to number grades.

Letter	Range
A	94-100
A-	90-93
B+	87-89
B	83-86
B-	80-82
C+	77-79
C	73-76
C-	70-72
D+	67-69
D	63-66
D-	60-62

F	< 60
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Mini-prototype (MP) Assignments

We'll have **three mini-prototype (MP) assignments** during the semester. For each MP, you'll have weekly instructions that help you start and develop that MP. These instructions do not have individual deadlines, except for the final instruction that completes each MP. The weekly instructions build toward each MP assignment, and the later MPs will be merged into one system for your final project. They exist to help you a) learn through firsthand experience, and b) achieve Learning Objectives 1-5.

How you'll be graded: Your work on each MP is graded through two components:

1. Weekly Status Update
2. MP Check-off Meeting

	1. Weekly Status update	2. Check-off meeting	Total points
MP 1	400	600	1000
MP 2	200	800	1000
MP 3	200	800	1000

1. Weekly Status Update (WSU)

- Each week, you are required to submit a brief status update (e.g. what did you do this week (1-3 sentences?)) via a Google Form linked on Canvas. This is worth 100 points per status. There will be 4 WSUs for MP 1 and 2 for MPs 2 and 3.
- Grading: This is graded pass/fail based on submission. It's okay to report a small update — you do not need a major milestone every single week, and it's fine to have no update for a single week. However, even if you do not have any updates, you still need to submit your WSU (complete the form) to receive points. Please note that you cannot have "no updates" in two consecutive weeks. If you didn't have any updates in your WSU this week, your next WSU must include some updates. If it does not, you will receive 0 points for that second week's status update. The content of your updates will be reviewed together, all at once, during your MP check-off meeting, and grades may be adjusted (deducted) after the check-off meeting. See below for more details.
- Late submissions: Not allowed.

2. MP Check-off Meeting

- Each MP is graded through a 10-minute check-off meeting with a TA. That means, after the deadline, you will need to meet with a TA.
- **Submission = your last commit before the deadline.** There is no separate submission step — whatever is in your GitHub repository at the MP deadline is treated as your submission. There will be no late submissions allowed. Make sure your final commit is in place before the deadline.

- **Scheduling your meeting:** TAs will reach out separately to arrange scheduling a week before each MP deadline. You must complete your 10-minute meeting with a TA within one week after the MP deadline. If you do not complete your meeting within this window, you will receive 0 points for that MP assignment.
- **No-shows and rescheduling:** If you miss your scheduled meeting or need to reschedule, you are allowed one reschedule, but the rescheduled meeting must still take place within the same one-week window after the MP deadline. If you miss or fail to attend a second time, or if the rescheduled meeting cannot happen within that one-week window, you will receive 0 points for that MP assignment.
- **What to expect in the meeting:** The TA will ask you to walk through or demonstrate parts of your implementation, and confirm that your code works as expected. When asked, you are expected to describe and explain your work. The TA will also compare your WSUs against your actual commit history. If these do not match – for example, your status updates describe progress that isn't reflected in your commits – your WSU grade may be deducted.
- More detailed information on grading rubric will be provided.

If you have a [Disability Access Services](#) accommodation related to assignments (e.g. extensions) and you'd like to use the accommodation for MP assignments, please let me know directly via email.

Please carefully read the Academic Integrity, External Resources, and Plagiarism and Cheating sections of the syllabus before working on your assignments.

Quiz

We'll have *in-class, on-paper, closed-book* quizzes in the beginning of every Tuesday class. The quiz is to help you acquire the ability to apply knowledge and skills without external resources.

- We will use the first 15 minutes of class to take the quiz.
 - **Quiz days and times**
 - Every Tuesday from Week 2 through Week 12. Please see the course schedule below.
 - Section 1 (1:35-3:15pm): 1:40pm-1:55pm.
 - Section 2 (3:25-5:05pm): 3:30pm-3:45pm.
 - Note: Quiz will begin and end promptly at the scheduled times.
- Each quiz will have 4-8 questions, all of which will be multiple-choice.
- Only pens, pencils, and erasers are allowed. Nothing else.

If you have a [Disability Access Services](#) accommodation related to quizzes/exams, make sure to schedule the accommodation at your earliest convenience – it is your responsibility to arrange to take the quizzes in the DAS office.

- Section 1 (1:35-3:15pm): the earliest time you may start the quiz at DAS is 1:30pm.
- Section 2 (3:25-5:05pm): the earliest time you may start the quiz at DAS is 3:20pm.
- Once you finish the quiz, please join the class.

Quizzes are worth 54% of your final grade, and they will be given 10 times throughout the semester.

However, there is no fixed number of quizzes you must take to reach full credit. Instead, every quiz you take contributes points toward your quiz grade, up to a maximum of 5400 points. How many quizzes it takes to reach that maximum depends on how well you score on each one.

Points Awarded per Quiz

Raw Score on Quiz	Points Earned
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95-100%	1450
80-94%	1100
61-79%	800
45-60%	540
25-44%	200
1-24%	50
0%	0

How Many Quizzes Are Needed for Full Credit (54%)?

Consistent Score Range	Quizzes Needed to Reach the full points (5400 pts)
95-100%	4
80-94%	5
61-79%	7
45-60%	10 (all of them)

Note:

- A student who consistently scores 95-100% only needs 4 quizzes to reach the maximum. In other words, you can lock in full credit with the minimum of 4 quizzes only – you do not need to grind through all 10.
- Missed or bad quizzes are not punished directly. Because points only accumulate toward a cap (rather than being averaged), a few weak or missed quizzes simply contribute fewer points toward the same 5400-point goal.
- Consistent effort is rewarded. A student, who scores in the 45-60% range every time, can still reach full credit.
- Extra effort is still rewarded, not wasted. See “What if I score more than 5400 in the end? (bonus-point opportunities)”.

No Practice Questions

No practice questions will be provided. Each quiz covers the lecture material from the previous week, so please use the lecture materials themselves to prepare. Within the lecture documents, some parts will be marked with “(quiz)”. Studying these marked parts specifically should be enough to get you into the 45-60% range on each quiz. In order to score above 60%, you will need to review the full lecture material, not just the marked sections.

No Make-Up Quizzes nor Averaging/Curving

Because this grading system is cumulative (points build up toward the 5400-point cap rather than being averaged), there are no make-up quizzes. If you need to miss a quiz, you miss it and you can take the next one – you do not need to notify the teaching team. You can decide whether to attend each quiz based on your own schedule and priorities.

Missing a few quizzes over the semester is okay and, depending on how you score on the remaining quizzes, may not hurt your overall grade. However, we do not recommend missing quizzes especially early in the semester. The earlier you fall behind in accumulating points, the fewer opportunities you have left to reach the full credit.

There is also no averaging or curving if you fall short of the 5400 points. Your quiz grade is simply the capped sum of the points you actually earned. Nothing will be scaled up or adjusted to compensate for missed or low-scoring quizzes. For example, if you take only two quizzes and earn 1450 points (100%) on each, your total is 2900/5400, and that will be your quiz grade.

What if I must miss one or more quizzes?

There may be times when you need to prioritize other responsibilities. To provide some flexibility, there will be an **optional quiz** during the week following Quiz 10. This quiz will not count as an 11th quiz. Instead, it can be used to replace *one* of your lowest quiz scores. For example, if you missed a quiz and received 0 points for it, you can take the optional quiz to replace that 0-point score.

What if I score more than 5400 in the end? (bonus-point opportunities)

Every additional 2000 points earned beyond 5400 converts into 2% bonus points on your final grade. So, if you earned 7400 points total, 5400 of those points are used for your 54% quiz grade, and the remaining 2000 points convert to an additional 2% on your final grade. You won't receive any bonus for excess points between 1 and 1999 – you need a full 2000-point increment to unlock each 2% bonus. For example, if you have 6000 points in total, you receive the full 54% quiz grade. The extra 600 points do not reach the 2000-point threshold, so no bonus is earned. If you have 10,000 points in total, your 4600 excess points contain two full 2000-point increments (the remaining 600 do not reach the next increment). Therefore, you receive the full 54% quiz grade plus a 4% bonus.

Semester Team Project

You'll work as a team (3-4 members) to complete a project over the course of the semester. The semester project is to help you acquire learning objects 1-5.

You'll choose a track and delve into it with your teammates. We expect a substantive project that incorporates the contributions of all teammates. Six separate components will be graded:

1. Team Report	100 pts
2. Progress Report	100 pts
3. Demo Testing	50 pts
4. Demo Presentation	250 pts
5. Final Deliverable	300 pts
6. Peer Evaluation	800 pts
Total	1600 pts

More details, such as each requirement and instruction, will be provided later in the semester. No late submissions will be accepted for the project-related submissions.

Bonus-point opportunities

See the section “What if I score more than 5400 in the end?” under the Quiz above.

Course Schedule

- Items that are due by that date or taking place on that day are **highlighted in bold**.
- Topics covered in each class may change while **bolded items** – deadlines, the quiz day, and the presentation days – **remain fixed**.

Week 1			Friday, 9/11, c1
			Introduction, Git set-up, Gradescope set-up, Git basics
Week 2	Monday, 9/14	Tuesday, 9/15, c2	Friday, 9/18, c3
	MP1-part 1 out	CLI Tools, argparse, Intro to Logging Quiz 1	Config Files & Project Organization MP1 Weekly Status Update (WSU) 1
Week 3	Monday, 9/21	Tuesday, 9/22, c4	Friday, 9/25, c5
	MP1-part 2 out	Filesystem Navigation with Pathlib & Data Formats Quiz 2	Git Branching MP1 WSU2
Week 4	Monday, 9/28	Tuesday, 9/29, c6	Friday, 10/2, c7
	MP1-part 3 out	Virtual Environments & Pandas Quiz 3	Filtering, Statistics & Transformations MP1 WSU3
Week 5	Monday, 10/5	Tuesday, 10/6, c8	Friday, 10/9, c9
	MP1-part 4 out	Missing Data & Outliers Quiz 4	Cleaning Pipelines & Text Cleaning with Regex MP1 WSU4
Week 6	Monday, 10/12	Tuesday, 10/13, c10	Friday, 10/16, c11
	MP 1	pytest Fundamentals & Fixtures Quiz 5	Testing MP1 & Coverage
Week 7	Monday, 10/19	Tuesday, 10/20, c12	Friday, 10/23, c13
	MP 2-part 1 out	FastAPI Fundamentals Quiz 6	Understanding Async in FastAPI MP2 WSU1
Week 8	Monday, 10/26	Tuesday, 10/27, c14	Friday, 10/30, c15
	MP 2-part 2 out	Middleware, Request IDs, Exception Logging & TestClient	MP2 Push Day

		Quiz 7	MP2 WSU2
Week 9	Monday, 11/2	Tuesday, 11/3, c16	Friday, 11/6, c17
	MP 2 MP 3-part 1 out	SQL, SQLite & Logging File Handlers Quiz 8	PostgreSQL, Snowflake Concepts & Testing Track (project) Overview MP3 WSU1
Week 10	Monday, 11/9	Tuesday, 11/10, c18	Friday, 11/13, c19
	MP 3-part 2 out	Introduction to Caching with Redis Quiz 9	Cache Invalidation, FastAPI Integration & Performance MP3 WSU2
Week 11	Monday, 11/16	Tuesday, 11/17, c20	Friday, 11/20, c21
	MP 3	GroupBy & Aggregations Quiz 10	Rolling Windows, Streamlit Project Team Report
Week 12	Monday, 11/23	Tuesday, 11/24, c22	Friday, 11/27, c23
		New Tools Introduction (track-specific: Dash/MLflow/DuckDB) Optional Quiz	Thanksgiving Day
Week 13	Monday, 11/30	Tuesday, 12/1, c24	Friday, 12/4, c25
		Working on your track - building core feature	Working on your track - building and testing Project Progress Report
Week 14	Monday, 12/7	Tuesday, 12/8, c26	Friday, 12/11, c27
		Project Demo Presentation 1 *Demo Testing is due one day before.	Project Demo Presentation 2 + Course Wrap-up *Demo Testing is due one day before.
Final Week	Monday, 12/14	Tuesday, 12/15	Friday, 12/18
	Project Final Deliverable Peer Evaluation		

* Demo testing is due one day before your presentation. For example, if you present on Tuesday, your testing submission is due Monday.

Reaching out to Instructor

As instructors of this course, we are here to support your learning journey in every way possible and to help you succeed in this course. We will do our best to support you. We plan to check in with you as needed, either before or after classes or via email. However, we may overlook your needs, so please don't just wait. Reach out to your lecture instructor via email for any concerns or obstacles hindering your learning or anything sensitive in nature. During the week, we'll do our best to respond within 24 hours, but don't expect a response after 5pm. On the weekends we'll be slower to respond, if you reach out over a weekend you can expect to hear back in the following week..

Your feedback is important for enhancing your learning experience and it will not impact your grades. You can reach out by speaking directly or sending an email.

Please come to office hours. If these times don't work, we can arrange a meeting. To schedule an alternative meeting time, email us with the course number in the subject line; otherwise, we may miss your email. We look forward to a productive and enjoyable learning experience together.

Academic Integrity

AI and External Resources

You may use external resources, including AI tools, books, websites, etc., to enhance your understanding of course material. If you use an external resource while completing coursework, you must work out the solution yourself and/or write your responses in your own words. AI tools may be used to explain concepts, help you understand errors, brainstorm ideas, or provide feedback on your work. However, you should not use AI to generate a solution or response that you submit as your own work. Using external resources beyond these purposes—including copying and pasting material or code and submitting it as your own—is considered plagiarism and cheating. You may also be asked to explain your work individually at any time. If you cannot explain how your code works or why you made particular implementation choices, it will be considered a violation of academic integrity.

For detailed information on what constitutes plagiarism and cheating and the associated consequences, refer to [Plagiarism and Cheating](#).

Collaboration

Collaboration is highly encouraged in this class, as a form of active learning. However, similar to AI and External Resources policy above, you must write your submission on your own, in your own words, before turning it in. The solution cannot be the same. If you worked with someone else, you must list all collaborators on your written submission to ensure proper credit is given. Copying solutions from a classmate is considered cheating. For detailed information on what constitutes these actions and the associated consequences, refer to [Plagiarism and Cheating](#).

Plagiarism and Cheating

Engaging in cheating or plagiarism will result in three things: 1) 0 points for the particular work, 2) a deduction to your final grade (equivalent to 10% of the final grade), and 3) a report filled with [OSCCR \(Office of Student Conduct and Conflict Resolution\)](#). The university's academic integrity policy discusses actions regarded as violations and consequences for students: <https://osccr.sites.northeastern.edu/academic-integrity-policy/>

- Submissions:
 - o All written submissions must be your own, original work. Original work for narrative questions is not mere paraphrasing of someone else's completed answer. You must not share written answers with each other at all. In other words, no two submissions should be identical or excessively similar, such as by only changing variable names. If two or more solutions appear very similar, even if not identical, the teaching team may contact the involved parties for further investigation.
 - o You may use AI tools (e.g., ChatGPT, Claude, or GitHub Copilot) only in ways that support your own learning and do not replace your own work. You are responsible for understanding, verifying, and being able to explain anything you submit. See AI and External Resources for more information.
- Quizzes: In addition to other forms of cheating, failure to follow the quiz instructions (e.g., continuing to work after time has expired, using anything else other than what's allowed, etc.) will be considered cheating.

Accommodation

Students who wish to receive academic services and/or accommodations should visit the Disability Access Services (DAS) at 20 Dodge Hall. For more information, visit <https://disabilityaccessservices.sites.northeastern.edu/>. Once you have done so, please provide your letter from the DAS to me as early in the semester as possible, and no later than one week before the date you require accommodations. I will arrange the necessary accommodations. Please note that accommodations may not be possible if the letter is provided too late.

For any other accommodations, please have [WeCare](#) contact me on your behalf. Once we get confirmation from them, we will figure it out.

Student athletes who anticipate challenges in being able to participate in class or submit assignments on time should speak to us as soon as possible about available alternatives or allowances.

Title IX Protections and Resources

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal funding. Any NU community member who has experienced such discrimination, sexual assault, relationship violence, stalking, coercion, and/or sexual harassment, is encouraged to seek help. Confidential support and guidance can be found through [University Health and Counseling Services](#), the Northeastern [Center for Spirituality, Dialogue, and Service](#), and the [Office of Prevention and Education at Northeastern \(OPEN\)](#).

Note that faculty members are considered “responsible employees” at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator.

Alleged violations can be reported non-confidentially to the Title IX Coordinator within The Office for Gender Equity and Compliance at: titleix@northeastern.edu and/or through NUPD (Emergency 617.373.3333; Non-Emergency 617.373.2121). Reporting Prohibited Offenses to NUPD does NOT commit the victim/affected party to future legal action. In case of an emergency, call campus police

or 911. For additional information and assistance please see the [Office of Institutional Diversity and Inclusion](#) webpage.

Wellness Day

As part of Northeastern's Wellness Days program, you may take a Wellness Day when needed – for more information, visit <https://wellnessdays.studentlife.northeastern.edu>. But please note that in-person assessments, such as quizzes and presentations, cannot be excused for Wellness Days. You will still need to attend and complete these assessments.

WeCare

[WeCare](#) is available to support students experiencing unexpected challenges that may affect their academic progress. If you are experiencing any challenges that are affecting your learning, please feel free to reach out to me and/or WeCare.